Simulating Materials Flow

Module 5 - ABE 721: Agricultural Systems Modelling and Simulation

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Lecture Outline

1. 1. Simulating Cyclical Systems
2. 2. Spatial and Temporal Models Integration
3. 3. Evaporation Models
4. 4. Crop Growth and Active Management Simulation

Introduction to Materials Flow

Agricultural systems are fundamentally thermodynamic and biogeochemical engines.

- They do not generate mass from nothing.
- They receive exogenous energy (solar radiation) to physically reorganize and transport materials (C, N, P, H_2O).
- Understanding the physics and biology requires explicitly simulating the **Flow of Materials**.

Compartment Modelling

In ecology, elements are continually recycled between the atmosphere, soil, microbes, and plants. We use **Compartment Modelling** to simulate this.

Conservation of Mass

The transient rate of change of mass in compartment i equals all inflows minus all outflows, plus boundary inputs (z_i) minus boundary exports (y_i):

$$\frac{dx_i}{dt} = \sum_{j \neq i} F_{ji} - \sum_{j \neq i} F_{ij} + z_i - y_i$$

Linear Donor-Controlled Systems

In most biogeochemical models, the flow rate F_{ij} is proportional to the mass currently in the source (donor) compartment i.

$$F_{ij} = a_{ij} x_i$$

- a_{ij} is the fractional transfer coefficient ($time^{-1}$).
- Represents the fraction of compartment i transferred to j per unit time.

Matrix Formulation:

$$\mathbf{x}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{z}(t)$$

Residence Time and Turnover Rates

Mean Residence Time (τ_i): The average duration a single atom (e.g., Carbon) spends inside compartment *i* before exiting.

In steady-state equilibrium ($\frac{dx_i}{dt} = 0$):

$$\tau_i = \frac{x_i^*}{F_{out,i}} = \frac{1}{|a_{ii}|}$$

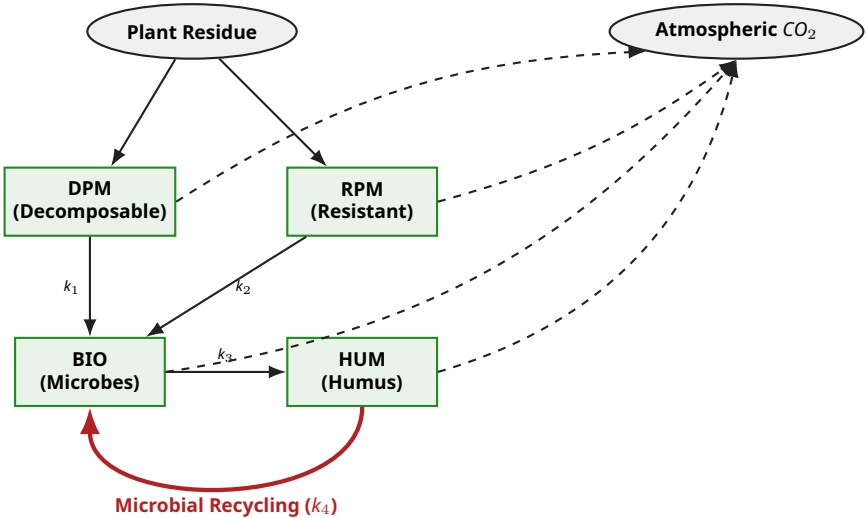
- **Turnover Rate:** The reciprocal ($1/\tau_i$).
- *Example:* Microbial biomass turnover is measured in days; humic organic matter turnover is measured in centuries.

Case Study: Soil Carbon Cycle (RothC)

The Rothamsted Carbon Model (RothC) divides soil carbon into compartments based on decomposition resistance:

1. **DPM:** Decomposable Plant Material (Fast turnover)
2. **RPM:** Resistant Plant Material (Medium turnover)
3. **BIO:** Microbial Biomass (Fast turnover recycler)
4. **HUM:** Humified Organic Matter (Slow turnover)

RothC Compartment Flow Diagram



Carbon Cycle ODEs

The cyclical nature arises because as microbes (*BIO*) decompose humus (*HUM*), they die and their mass cycles backward to become new organic input.

Microbial Biomass ODE

$$\frac{d(BIO)}{dt} = a \cdot k_1(DPM) + b \cdot k_2(RPM) + c \cdot k_4(HUM) - k_3(BIO)$$

Where k_i are specific rate constants, and a, b, c represent the fraction of carbon assimilated by microbes rather than respired instantly as CO_2.

Spatio-Temporal Transport

Lumped (0D) models treat a farm as a perfectly mixed bucket.

- Real physical flows (leaching nitrate, spreading pests) occur across **geographic space** and **continuous time**.
- Simulating these materials explicitly requires the integration of spatial algorithms and temporal calculus via Partial Differential Equations (PDEs).

The Advection-Dispersion-Reaction Equation

The movement of a dissolved material (solute) through porous soil is governed by the **ADRE**.

Three simultaneous physical processes:

1. **Advection:** Transport due to bulk fluid flow ($J_a = v \cdot C$).
2. **Dispersion:** Transport due to microscopic gradients and tortuous mixing ($J_d = -D\frac{\partial C}{\partial x}$).
3. **Reaction:** Biological/chemical decay or transformation ($R = -\mu C$).

Derivation of the 1D ADRE

Applying Conservation of Mass across the geometric boundaries yields the transient 1D ADRE:

1D Advection-Dispersion-Reaction Equation

$$\frac{\partial C}{\partial t} = \underbrace{D\frac{\partial^2 C}{\partial x^2}}_{\text{Dispersion}} - \underbrace{v\frac{\partial C}{\partial x}}_{\text{Advection}} - \underbrace{\mu C}_{\text{Reaction}}$$

The 2D ADRE Extension

For 2D simulations (e.g., fertilizer leaching downwards into aquifers and laterally into adjacent streams), the equation expands into multivariate calculus:

$$\frac{\partial C}{\partial t} = \left[D_x \frac{\partial^2 C}{\partial x^2} + D_z \frac{\partial^2 C}{\partial z^2} \right] - \left[v_x \frac{\partial C}{\partial x} + v_z \frac{\partial C}{\partial z} \right] - \mu C$$

How do we computationally solve this without the simulation mathematically exploding?

Numerical Integration: The ADI Method

Standard Explicit FDM suffers from severe **CFL stability constraints** when handling advection-dominated flows. High velocities (v) cause mathematical explosion.

- Solution: **Alternating Direction Implicit (ADI) Method**.
- Unconditionally stable.
- Splits a massive 2D PDE grid into two simpler 1D problems.
- Alternates the direction of implicit calculation every half time-step.

ADI Method: X-Sweep and Z-Sweep

Step 1 (X-Sweep): Advance simulation to $t + \Delta t/2$.

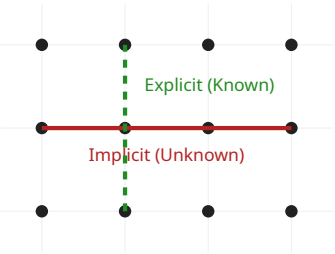
- Calculate X-derivatives *implicitly* (unknowns).
- Calculate Z-derivatives *explicitly* (knowns).
- Solves horizontal rows instantly using the Thomas Algorithm.

Step 2 (Z-Sweep): Advance to $t + \Delta t$.

- Calculate Z-derivatives *implicitly*.
- Calculate X-derivatives *explicitly* using Step 1 results.
- Solves vertical columns.

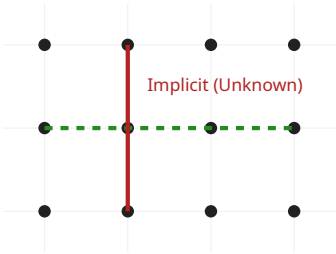
ADI Grid Integration

Step 1: X-Sweep ($t + \Delta t/2$)



Advance $\Delta t/2$

Step 2: Z-Sweep ($t + \Delta t$)



Evaporation and Atmospheric Flows

The most massive and energetic material transfer occurring on any farm is the continuous phase-change flow of liquid water into atmospheric vapor.

Modelling Evaporation paradigms:

1. Mass Transfer (Aerodynamic) Models
2. Energy Balance Models
3. Combination Models (Penman-Monteith)

1. Mass Transfer (Aerodynamic) Models

Treats evaporation strictly as an aerodynamic diffusion process driven by Vapor Pressure Deficit (VPD).

$$E = f(u) \cdot (e_s - e_a)$$

Formalized using **Aerodynamic Resistance** (r_a), integrated logarithmically based on wind speed (u_z):

$$r_a = \frac{\ln\left(\frac{z-d}{z_{om}}\right) \ln\left(\frac{z-d}{z_{oh}}\right)}{k^2 u_z}$$

Rough canopies (sugarcane) create turbulence, lowering r_a and accelerating evaporation.

2. Energy Balance Models

Water requires massive thermal energy to break hydrogen bonds. The latent heat of vaporization (λ) ≈ 2.45 MJ/kg.

First Law of Thermodynamics mandates:

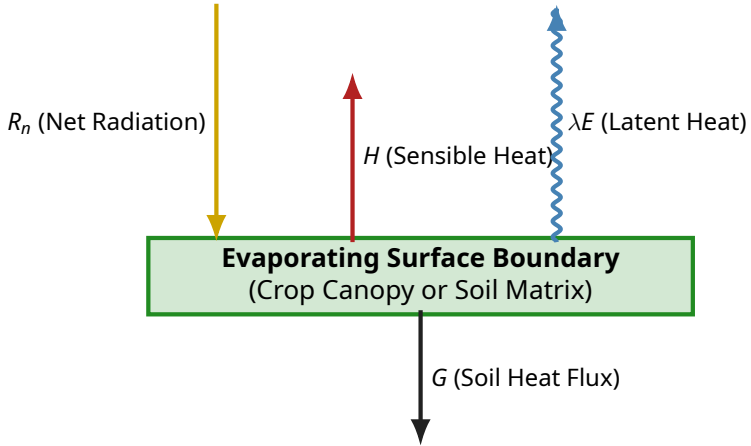
$$R_n = H + \lambda E + G$$

Using the **Bowen Ratio** ($\beta = H/\lambda E$), we eliminate sensible heat (H):

$$\lambda E = \frac{R_n - G}{1 + \beta}$$

Theoretical maximum evaporation is strictly dictated by available solar energy.

Thermodynamic Energy Balance



Conservation: $R_n - G - H - \lambda E \approx 0$

3. The Combination Method: Penman-Monteith

Aerodynamic models fail when radiation is low; Energy Balance models fail during advection (oasis effect).

The **FAO-56 Penman-Monteith Equation** combines both physics and adds biological **Canopy Surface Resistance** (r_s):

Penman-Monteith

$$\lambda ET_0 = \frac{\Delta(R_n - G) + \rho_a c_p \frac{(e_s - e_a)}{r_a}}{\Delta + \gamma\left(1 + \frac{r_s}{r_a}\right)}$$

As soil dries, stomata close, $r_s \to \infty$, and evaporation violently halts.

Resource-Limited Biomass Accumulation

Real-world growth is limited by the most deficient material flow, adhering to **Liebig's Law of the Minimum**.

Actual daily growth rate is the potential maximum multiplied by a dimensionless scalar stress factor ($0 \leq \mathit{Stress} \leq 1$):

$$\left.\frac{dW}{dt}\right|_{act} = \left.\frac{dW}{dt}\right|_{pot} \times \min(F_{water}, F_{nitrogen}, F_{temp})$$

Water Stress Factor (F_{water})

Calculated dynamically based on the ratio of actual transpiration (T_{act}) to potential transpiration (T_{pot}):

$$F_{water} = \frac{T_{act}}{T_{pot}}$$

- When soil dries, hydraulic resistance skyrockets.
- T_{act} drops below T_{pot}.
- F_{water} drops below 1.0.
- This instantly cripples the $\frac{dW}{dt}$ calculus, accurately simulating drought stunting.

Simulating Management: Discrete Impulses

How do we simulate a tractor applying fertilizer on precisely Day 45 within a continuous ODE simulation?

- Management events are modeled as **Dirac Delta Functions** ($\delta(t)$) or instantaneous state-resets.
- They are *discrete events* explicitly interrupting a *continuous simulation*.

Simulating Irrigation Events

An irrigation event at t_{irrig} with depth I_{depth} is simulated as an instantaneous mathematical jump condition for soil moisture (θ):

Instantaneous Jump

$$\theta(t_{irrig}^{+}) = \theta(t_{irrig}^{-}) + \frac{I_{depth}}{Z_{root}}$$

If the new θ exceeds Field Capacity (FC), the excess material is instantly shunted into deep drainage algorithms (runoff).

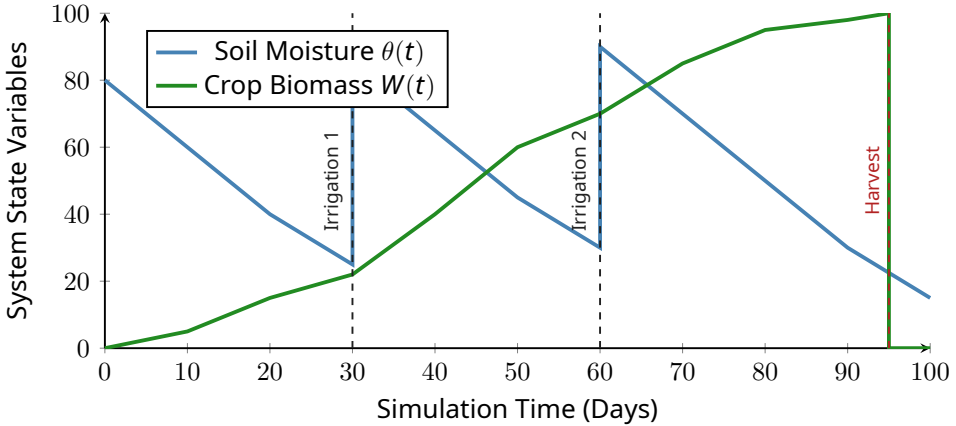
Simulating Harvest Events

Harvesting is the ultimate disruption of material flow.

- Accumulated grain biomass is removed (Yield).
- Vegetative biomass is instantly transferred into the *DPM* (Decomposable Plant Material) compartment.
- The biological state is reset to zero.

$$
\begin{aligned}
Y_{harvest} &= W_{grain}(t^-_{harvest}) \\
DPM(t^+_{harvest}) &= DPM(t^-_{harvest}) + W_{leaf/stem}(t^-_{harvest}) \\
W_{total}(t^+_{harvest}) &= 0
\end{aligned}
$$

Hybrid Continuous-Discrete Simulation



Summary of Module 5

- **Cyclical Systems:** Modelled using mass-balance compartments and linear donor-controlled matrices (e.g., RothC).
- **Spatio-Temporal Integration:** Required the ADRE solved via the unconditionally stable ADI computational method.
- **Evaporation:** Penman-Monteith unifies aerodynamic mass transfer and thermodynamic energy balance.
- **Management Simulation:** Hybrid models use Dirac Delta functions to impose discrete human events on continuous biological ODEs.

End of Module 5

Questions & Discussion